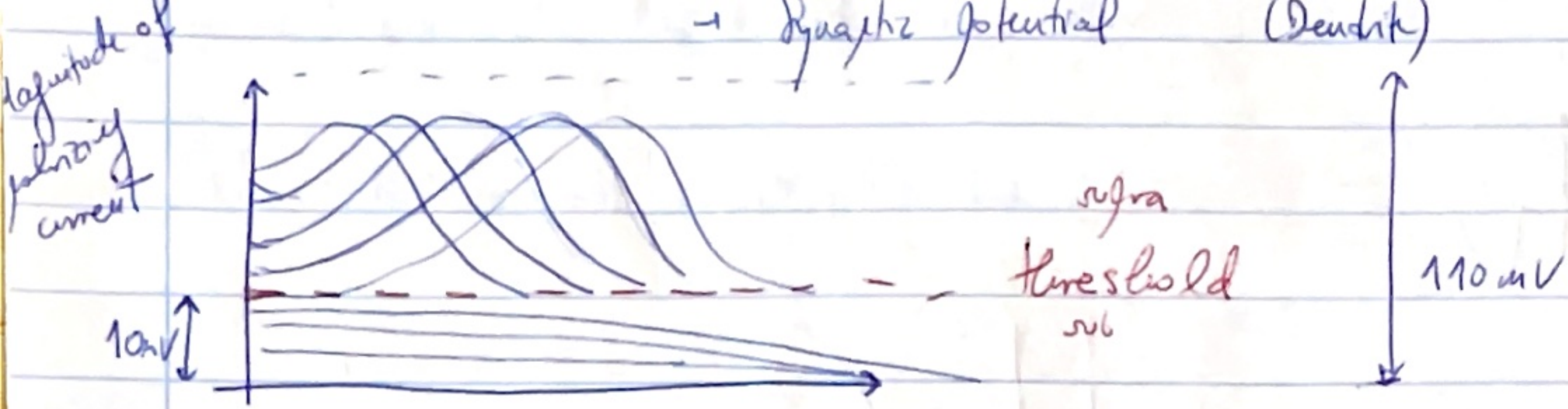
WEEK 4

Spike → all or none → Action potential (Axon)
→ Synaptic potential (Dendrite)



The spikes are all the same ↱ suprathreshold is all (1)
subthreshold is none (0)

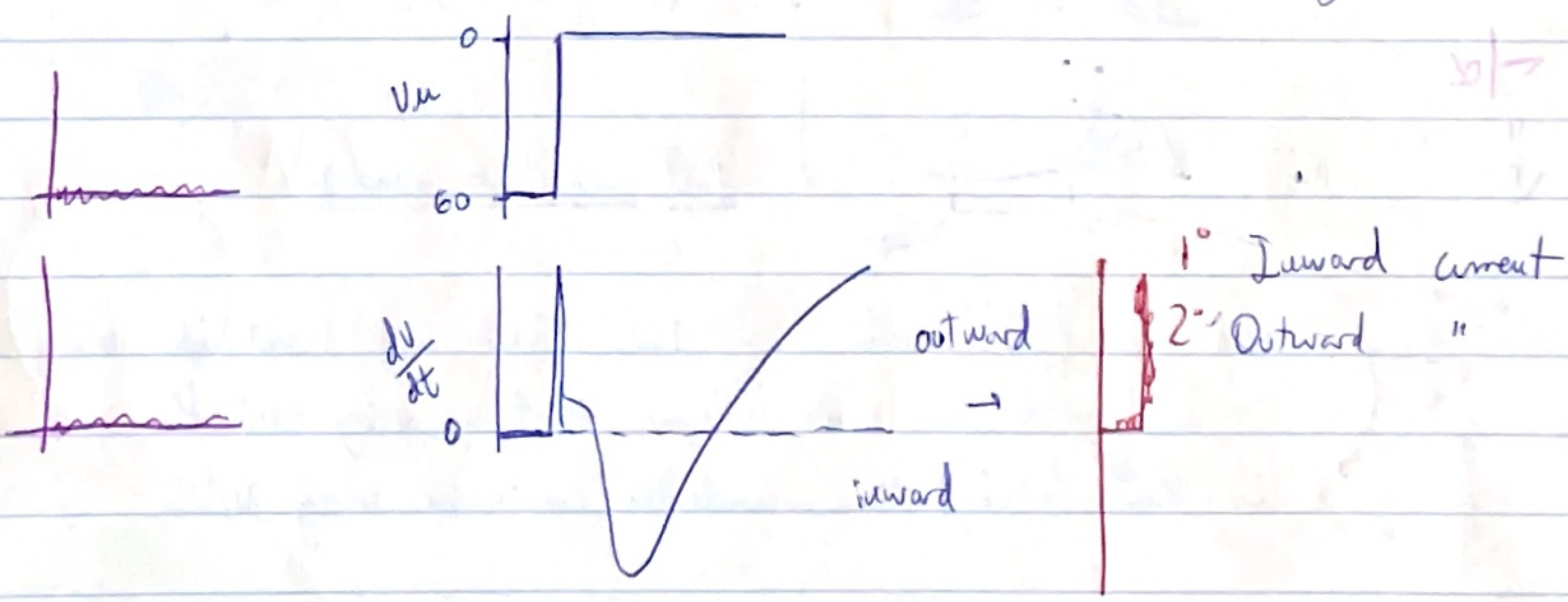
Hodkin - Huxley.

What is unique in the axonal membrane to produce this all/none potential?

They put a resistance to make the membrane isopotential
Space CLAMP
↳ because of axial resistivity the "space is reduced"

VOLTAGE CLAMP
Fix the voltage both inside and outside (I don't want the action potential to interfere)

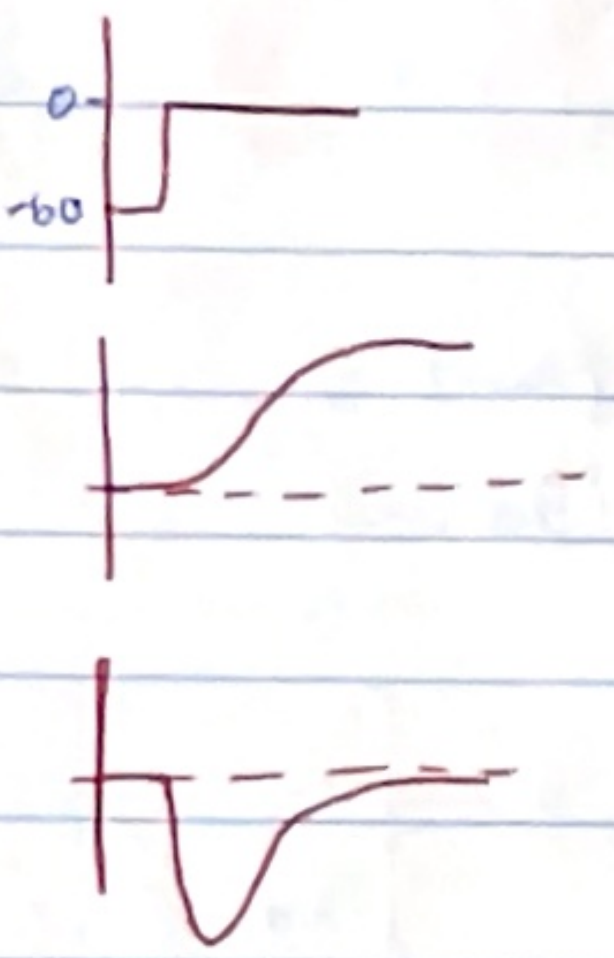
what are the current that flows through the membrane for a given voltage?



Capacitance = charge (Q) / Volts (V)   Farads

→ TTX

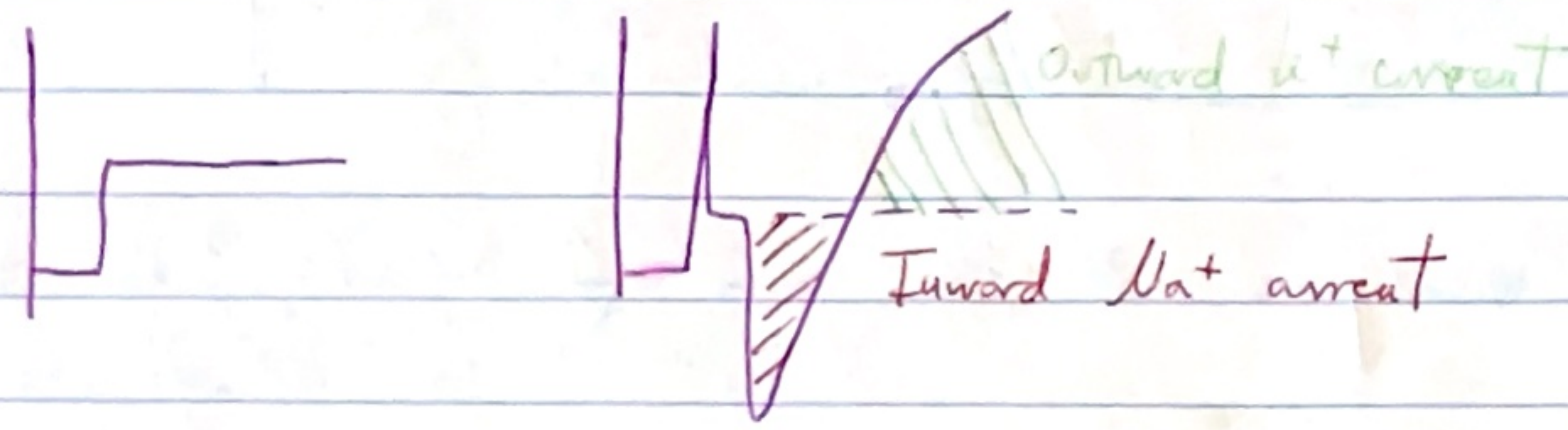
→ TEA

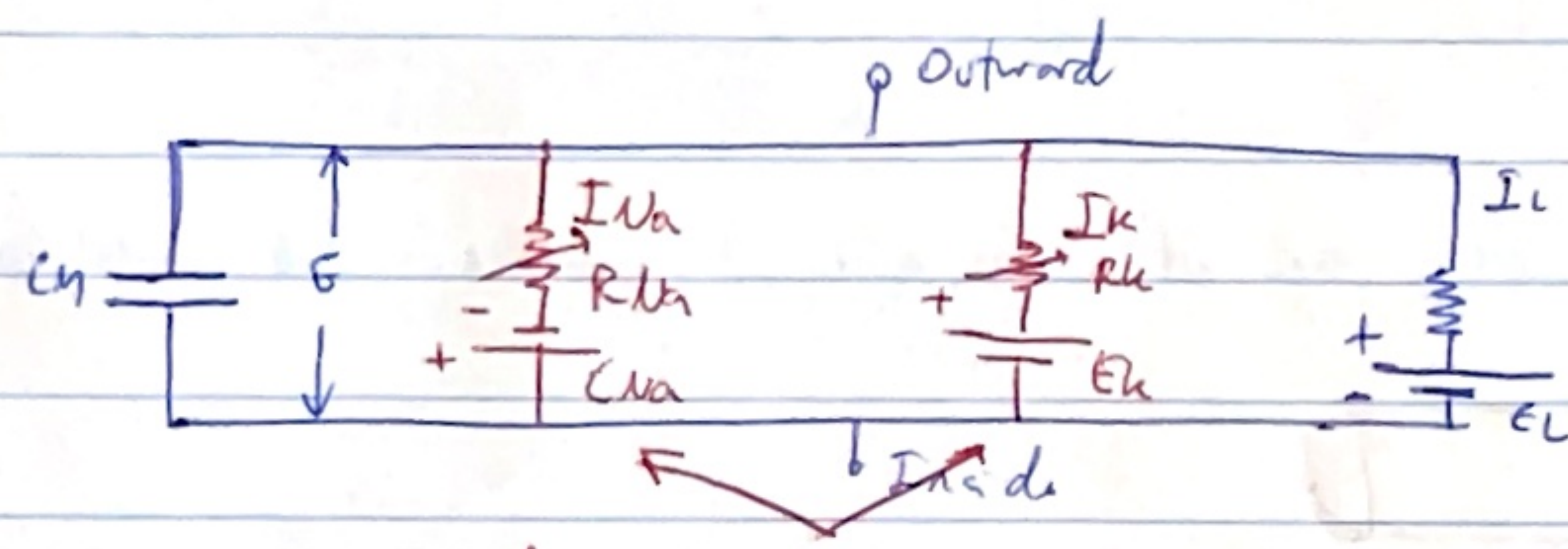


- Using drugs they started to separate the currents

With TTX they erased the inward current

With TEA they erased the outward current





L = leak (passive

Voltage dependent

$$I = C_M \dot{V} + g_{Na}\, h m^3 (V - V_{Na}) + g_K\, n^4 (V - V_K) + G_L (V - V_L)$$

Grand equation

• Circuitos

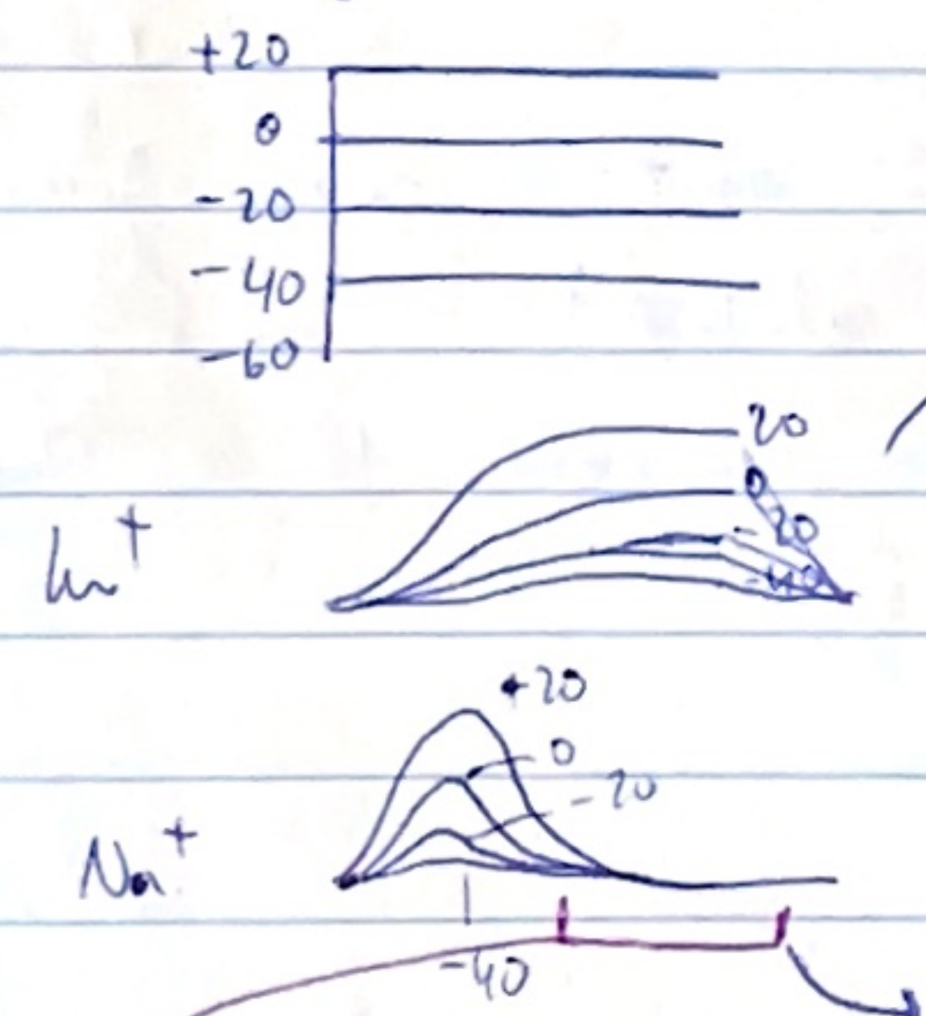
Ohm: $V = R \times I$ — Voltios diversos ... anterior

$G = \frac{1}{R}$ facilidad para paso de corriente

R = Resistencia a la corriente

Modeling the membrane Currents

$I_K = g_K (V_M - E_K)$



It takes time to respond (later current) slow conductance change

Voltage dependent conductance

fast conductance change

- The slow (K) current (conductance) does not inactivate during VC
- The K conductance rises slower than it decays at the end of VC
- The fast (early) Na conductance inactivates during VC

$g_k = \bar{g}_k \cdot n^4$ t dependent

↳ V dependant, 0-1. (~% open channels)

$\bar{g}_k$ ↓ maximum possible conductance

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⇒ There is a refractory period

absolute refractory period = Na^+ inactivation

K^+ activation (slow)

WEEK 5

"Hebb hypothesis"

"Fire together wire together"

Synaptic plasticity

FUNCTIONAL PLASTICITY

Spike timing-dependent synaptic plasticity (STDP)

long term potentiation (LTP) → ~~more time~~ the longer a synapse occurs, the stronger it gets

↳ LTD = if you stimulate first the post synaptic neuron then the synapse gets weaker (depression)

↙

milliseconds mechanism

STRUCTURAL PLASTICITY

Globus & Scheibel – Sensory experience (visual deprivation) affects spine variation in rabbits

- New dendritic spines are born constantly
- More often so during learning tasks / enriched environment
- New spines are associated with new synapses – new functional networks.

NEUROGENESIS & LEARNING

1997 Elizabeth Gould → neurogenesis
At least 2 regions that constantly generate newborn cells.

- Prof Adi Mizrahi lab → watching newborn cells growing in the adult brain
  hippocampus → associated with certain types of behaviour.
  ↓
  they grow and mature
- they are born as stem cells.

The most you use the network, the more connections, the more synapses, the more stem cells...

- Taxi drivers London → quantity / volume in hippocampus & years of experience.

The future

- Brain inspired learning machines
- How trustworthy are your memories?
- Could we read out memories? (No, everyone codes in a ≠ way)
- Could we embed new memories?